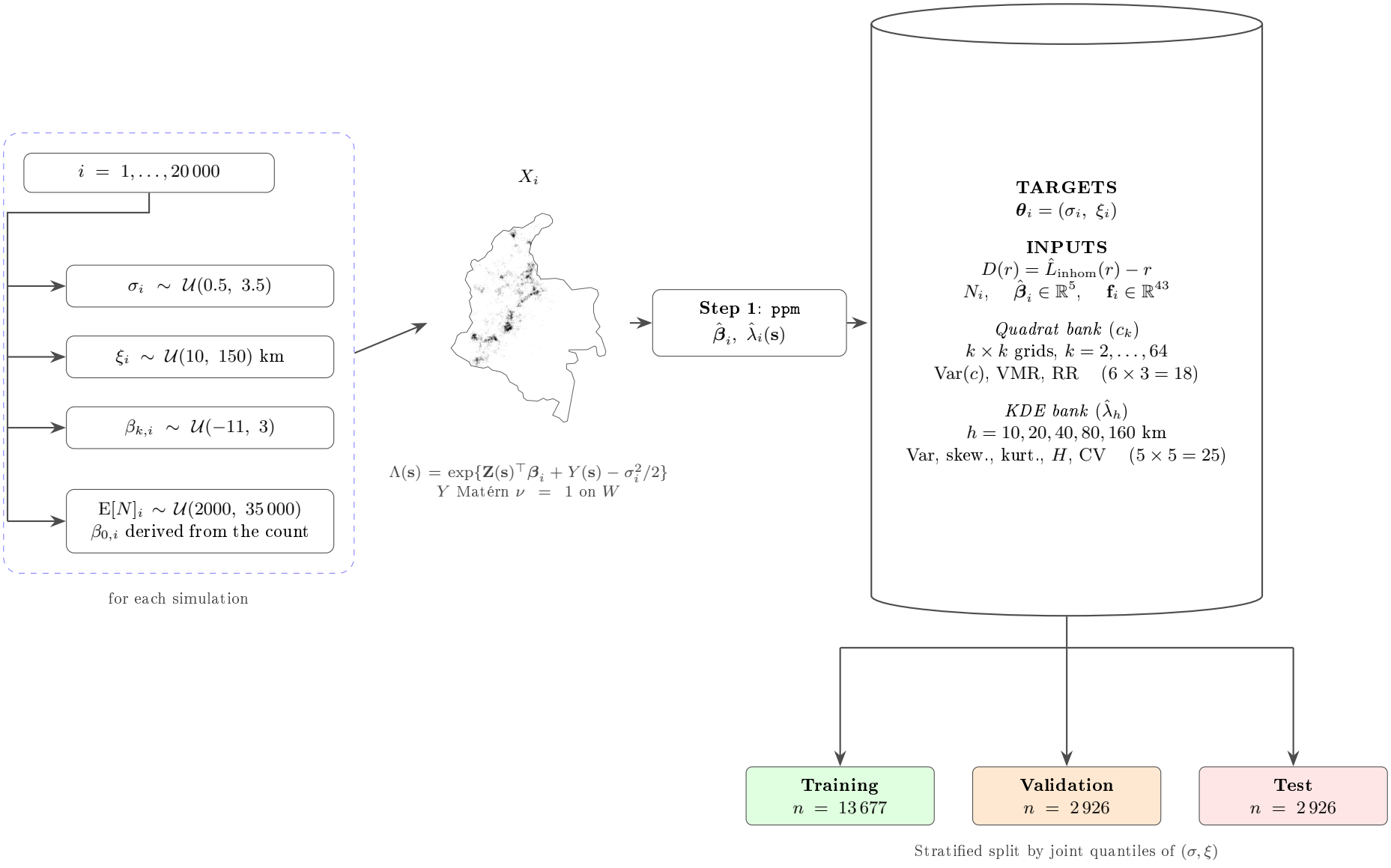

$i = 1, \ldots, 20\,000$
$\sigma_i \sim \mathcal{U}(0.5,\ 3.5)$
$\xi_i \sim \mathcal{U}(10,\ 150)$ km
$\beta_{k,i} \sim \mathcal{U}(-11,\ 3)$
$\mathrm{E}[N]_i \sim \mathcal{U}(2000,\ 35\,000)$
$\beta_{0,i}$ derived from the count
for each simulation
X_i
$\Lambda(\mathbf{s}) = \exp\{\mathbf{Z}(\mathbf{s})^\top \boldsymbol{\beta}_i + Y(\mathbf{s}) - \sigma_i^2/2\}$
Y Matérn $\nu = 1$ on W
Step 1: ppm
$\hat{\boldsymbol{\beta}}_i,\ \hat{\lambda}_i(\mathbf{s})$
TARGETS
$\boldsymbol{\theta}_i = (\sigma_i,\ \xi_i)$
INPUTS
$D(r) = \hat{L}_{\mathrm{inhom}}(r) - r$
$N_i, \quad \hat{\boldsymbol{\beta}}_i \in \mathbb{R}^5, \quad \mathbf{f}_i \in \mathbb{R}^{43}$
Quadrat bank (c_k)
$k \times k$ grids, $k = 2, \ldots, 64$
Var(c), VMR, RR $\quad (6 \times 3 = 18)$
KDE bank ($\hat{\lambda}_h$)
$h = 10, 20, 40, 80, 160$ km
Var, skew., kurt., H, CV $\quad (5 \times 5 = 25)$
Training
$n = 13\,677$
Validation
$n = 2\,926$
Test
$n = 2\,926$
Stratified split by joint quantiles of (σ, ξ)